

DUMFRIES DRUNGANS, United KingdomWMO: **031540**Lat: **55.050N**Lon: **3.650W**Elev: **16**StdP: **101.13**Time Zone: **0.00 (EUW)**Period: **94-04**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-5.0	-3.0	-6.4	2.2	-3.0	-4.9	2.5	-1.2	11.9	8.9	10.3	8.3	0.1	130	0.559

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	7.7	24.4	18.3	22.2	17.2	20.3	16.3	19.2	23.4	17.8	21.2	16.7	19.5	2.3	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
17.4	12.5	21.4	16.3	11.6	19.8	15.5	11.0	18.4	54.8	23.5	50.4	21.4	47.0	19.8	25.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
9.4	8.2	7.0	-8.4	26.4	3.3	2.0	-10.8	27.8	-12.8	29.0	-14.6	30.1	-17.1	31.6		
			-8.7	21.5	3.3	1.7	-11.1	22.7	-13.1	23.7	-15.0	24.6	-17.4	25.8		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	9.3	4.3	4.8	6.2	7.8	11.0	13.1	15.2	15.5	12.9	9.9	7.0	4.0
	DBStd	4.93	3.00	2.94	2.66	2.65	2.83	2.32	2.37	2.88	2.17	3.01	2.94	3.85
	HDD10.0	871	178	146	121	75	22	3	1	1	3	40	97	187
	HDD18.3	3302	436	378	377	317	228	158	102	98	162	263	341	444
	CDD10.0	624	1	1	2	8	52	96	161	170	91	36	6	1
	CDD18.3	13	0	0	0	0	0	1	3	9	0	0	0	0
	CDH23.3	107	0	0	0	0	1	12	32	62	0	0	0	0
	CDH26.7	11	0	0	0	0	0	1	2	8	0	0	0	0
Wind	WSAvg	2.6	3.0	3.5	3.2	2.6	2.6	2.5	2.1	1.9	1.9	2.7	2.5	2.6
Precipitation	PrecAvg	1050	108	83	78	56	71	64	72	90	100	120	105	106
	PrecMax	1380	231	243	170	119	168	117	196	206	210	248	224	207
	PrecMin	697	11	3	26	6	7	10	23	13	10	22	18	32
	PrecStd	136	54	53	38	29	37	27	36	46	49	60	51	46
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.8	11.9	13.5	19.1	22.7	25.2	26.4	27.7	21.1	17.5	14.5	13.1
		MCWB	10.5	10.5	10.0	13.5	17.8	17.5	19.4	20.0	17.1	14.5	13.3	12.3
	2%	DB	10.3	10.9	12.0	15.1	20.6	21.4	23.8	24.9	19.2	16.0	13.2	11.1
		MCWB	9.0	9.6	9.7	11.7	15.5	16.2	17.8	19.4	15.4	14.0	11.9	10.0
	5%	DB	9.6	10.1	11.1	13.2	18.2	19.1	21.4	22.6	18.0	15.2	12.5	10.1
		MCWB	8.3	8.8	9.2	10.6	14.1	14.7	16.7	18.0	14.7	13.6	11.3	9.2
	10%	DB	8.8	9.2	10.2	12.0	15.8	17.2	19.5	20.3	16.9	14.5	11.6	9.2
		MCWB	7.6	7.9	8.4	9.8	12.5	13.5	15.8	16.7	14.3	12.9	10.5	8.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.7	10.9	11.6	14.2	19.1	18.8	20.3	20.8	17.7	15.2	13.7	12.3
		MCDB	11.5	11.5	12.1	17.3	21.4	23.9	24.8	26.4	20.4	16.3	14.4	13.0
	2%	WB	9.5	9.9	10.4	12.3	16.1	16.6	18.6	19.6	16.3	14.6	12.2	10.2
		MCDB	10.3	10.6	11.3	14.3	19.8	20.7	22.5	24.1	18.2	15.6	12.9	10.9
	5%	WB	8.5	9.1	9.6	11.2	14.4	15.2	17.2	18.3	15.4	13.9	11.4	9.3
		MCDB	9.4	10.0	10.7	12.7	17.5	17.9	20.4	21.9	17.1	15.0	12.3	10.1
	10%	WB	7.8	8.1	8.8	10.2	12.9	14.2	16.1	17.1	14.6	13.1	10.6	8.3
		MCDB	8.7	9.0	9.9	11.6	15.4	16.5	18.9	19.8	16.3	14.4	11.6	9.2
Mean Daily Temperature Range	5% DB	MDBR	4.5	5.1	5.8	6.8	7.4	6.7	6.8	7.7	7.0	6.2	5.1	4.7
		MCDBR	4.8	4.8	6.6	9.1	11.5	10.7	10.9	11.3	9.0	6.1	5.1	5.1
		MCWBR	4.6	4.4	4.9	6.5	7.4	6.5	6.3	6.4	5.8	4.8	4.9	4.9
	5% WB	MCDBR	4.6	4.4	5.5	7.9	10.5	9.1	9.5	10.2	7.4	5.8	4.9	5.1
		MCWBR	4.7	4.4	4.7	6.3	7.6	6.3	6.2	6.4	5.7	5.1	5.0	5.1
Clear-Sky Solar Irradiance	taub	0.328	0.339	0.365	0.384	0.387	0.384	0.385	0.371	0.360	0.344	0.321	0.313	
	taud	2.248	2.336	2.301	2.278	2.317	2.355	2.357	2.406	2.441	2.425	2.331	2.229	
	Ebn at Noon	583	734	800	837	856	862	853	844	803	722	596	507	
	Edh at Noon	59	77	99	116	118	115	113	101	86	70	54	48	
All-Sky Solar Radiation	RadAvg	0.55	1.20	2.29	3.68	4.78	4.73	4.55	3.69	2.62	1.43	0.70	0.41	
	RadStd	0.07	0.13	0.22	0.37	0.37	0.47	0.48	0.28	0.22	0.17	0.07	0.04	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (57 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page